BrightLight Tutorials – Data Analytics

SQL Fundamentals

Exercise 3: SQL CASE Statements

Instructions:

1. Write your answers on paper using a pen.
2. For each query, **draw a table** showing the final output (the result set).
3. In your `SELECT` statements, choose **relevant columns to display**, unless specified.
4. After completing all questions, **scan your work into a PDF**.
5. Email your PDF to: **rofhiwa@brightlighttutorials.co.za**
6. **Submission Deadline: 21 April 2025, 23:59**

Questions

1. Table: `products`

product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Question:

Classify each product by price:

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Expected Output Columns:

- `product_name`
- `price`
- `price_category`

2. Table: `orders`

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Question:

Label each order:

- 'High Value' for orders ≥ 1000
- 'Medium Value' for 500–999.99
- 'Low Value' for orders < 500

Expected Output Columns:

- `customer_name`
- `amount`
- `order_value_category`

3. Table: `employees`

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Question:

Categorize employee position:

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

Expected Output Columns:

- `emp_name`
- `department`
- `salary`
- `position_level`

4. Table: `students`

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Question:

Assign a letter grade:

- ≥ 90: 'A'
- 80–89: 'B'
- 70–79: 'C'
- 60–69: 'D'
- < 60: 'F'

Expected Output Columns:

- `student_name`
- `score`
- `grade`

5. Table: `deliveries`

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Question:

Label delivery performance:

- ≤ 30 mins: 'Fast'
- 31–60 mins: 'On Time'
- 60 mins: 'Late'

Expected Output Columns:

- `delivery_id`
- `delivery_time_minutes`
- `performance`

6. Table: `tickets`

ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

Question:

Convert priority to labels:

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

Expected Output Columns:

- `issue_type`
- `priority`
- `priority_label`

7. Table: `attendance`

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Question:

Calculate attendance % and classify:

- ≥ 90% → 'Excellent'
- 75–89% → 'Good'
- < 75% → 'Needs Improvement'

Expected Output Columns:

- `student_id`
- `attendance_percentage`
- `attendance_status`

8. Table: `products_inventory`

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Question:

Label stock status:

- 0 → 'Out of Stock'
- 1–5 → 'Low Stock'
- 5 → 'In Stock'

Expected Output Columns:

- `product_id`
- `stock_qty`
- `stock_status`

9. Table: `classes`

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Question:

Classify by size:

- ≥ 25 → 'Large'
- 10–24 → 'Medium'
- < 10 → 'Small'

Expected Output Columns:

- `subject`
- `enrolled_students`
- `class_size_category`

10. Table: `payments`

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

Question:

Apply discount flag:

- If `payment_method = 'Cash'` and amount ≥ 200 → 'Eligible for Discount'
- Otherwise → 'Not Eligible'

Expected Output Columns:

- `payment_id`
- `payment_method`
- `amount`
- `discount_eligibility`

SQL CASE STATEMENTS

Q1 Product Price Categories

```
SELECT
      Product_name,
      Price ,
  CASE
      WHEN price > 1000 THEN 'expensive'
      WHEN price BETWEEN 100 AND 1000 THEN 'mid-range'
      ELSE 'Budget'
      END AS price category
FROM products;
```

Product_name	Price	price_category
Laptop	1200.00	Expensive
Phone	800.00	mid-range
keyboard	45.00	Budget
Monitor	300.00	Midrange
Mouse	25.00	Budget.

Q2 ORDER value Categories

```
SELECT
  customer_name,
  amount ,
CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount >= 500 THEN 'Medium Value'
  ELSE 'Low value'
END AS order_value category
FROM ORDERS ;
```

customer_name	amount	order_value_category
ALICE	150.00	Low value
Bob	560.00	Medium Value
Charlie	999.99	Medium value
Diana	45.50	low Value
Ethan	1200.00	high value

Q3 Employee Position level

```
SELECT
        ~~employ~~
        emp-name,
        department,
        Salary ,
          CASE
             WHEN department = 'IT' AND salary > 80 000 THEN 'Senior IT'
             WHEN department = 'HR' AND Salary > 55000 THEN 'Experienced HR''
        ELSE AS position level
        From employees;
```

emp-name	department	salary	Position level
John	IT	85000	~~85 000~~ Senior IT
Sara	HR	60 000	experienced HR
Mark	IT	75 000	Staff
Lucy	finance	95 000	Staff
Tom	HR	55 000	Staff

Q4 STUDENT LETTER Grades

```
SELECT
    Student-name,
    Score ,
  CASE
      WHEN Score >= 90 THEN 'A'
      WHEN Score >= 80 THEN 'B'
      WHEN Score >= 70 THEN 'C'
      WHEN Score >= 60 THEND 'D'
  END AS grade
  From Students;
```

Student-name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Q5 Delivery Perfemance

```
SELECT
    delivery-id ,
    delivery - time - minutes ,
    CASE
        WHEN delivery-time - minutes <= 30 THEN 'Fast)
        WHEN delivery-time - minutes <= 60 THEN 'ONTime'
    ELSE 'Late'
  END AS perfemance
  FROM deliveries;
```

delivery - Id	delivery - time - minutes	Perfemance
1	45	onTime
2	80	LAte
3	30	Fast
4	65	late
5	100	late

Q6 Ticket Priority labels

```
SELECT
    issue - type ,
    Priority ,
    CASE WHEN priority = 3 THEN 'High'
         WHEN priority = 2 THEN 'medium'
         WHEN priority = 1 THEN 'Low
  END AS priority label
FROM tickets ;
```

issue - type	Priority - '	priority - label
login issue	1	low
Server down	3	High
Slow system	2	medium
email error	2	medium
Password reset	1	Low

Q7 Attendance Percentage & Status

```
SELECT
    Student_id,
  (days_Present * 100.0 /total days) AS attendance_percentage,
 CASE  WHEN (days_present * 100.0 / total_days >= 90  THEN 'Excellent'
        WHEN (days_Present * 100.0 / total days >= 75  THEN 'Good'
   ELSE 'needs_improvement'
  END AS attendance_status
  FROM attendance;
```

Student id	attendance_Percentage	Attendance_Status
1	90 %	Excellent
2	60 %	Needs Improvement
3	96 %	Excellent
4	50 %	Needs improvement
5	100 %	Excellent

Q8 Inventory stock status

```
  SELECT
     Product_id,
     stock qty,
      CASE
          WHEN  stock_qty = 0  THEN 'out of stock'
          WHEN  Stock_qty  BETWEEN 1 AND 5  THEN  'Low Stock'
    ELSE  'In stock'
  END AS   stock_status
  FROM  products_inventory;
```

Product_id	Stock_qty	Stock_status
1	5	low stock
2	0	out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

Q9 Class size Categories

```
SELECT
    subject,
    enrolled_students,
    CASE
        WHEN enrolled students >= 25 THEN 'Large'
        WHEN enrolled students BETWEEN 10 AND 24 THEN 'medium'
    ELSE 'small'
  END AS class_size_category
FROM classes;
```

Subject	Enrolled_students	class_size_category
maths	30	large
English	25	large
Science	15	Medium
Art	5	Small
History	20	medium

Q10 Payment Discount Eligibility

```
SELECT
    Payment_id,
    payment_method,
    amount,
    CASE
        WHEN paymen_method = 'cash' AND amount >= 200
        THEN 'Eligible for Discount'
    ELSE 'Not Eligible
  END AS discount_eligibility
FROM payments;
```

Payment_id	Payment_method	amount	discount_eligibility
1	card	50.00	Not Eligible
2	cash	200.00	Eligible for discount
3	card	150.00	Not eligible
4	Paypal	75.00	Not eligible
5	Cash	300.00	Eligible for Discount